



Volume of Solids with Known Cross Sections

Calculus Worksheet · Grade 11-12

Name: _____

Date: _____

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Learning Objectives

- Set up integrals to compute the volume of a solid with known cross sections
- Express the side length of a cross section using the base region's equation
- Evaluate definite integrals to find exact volumes of solids

For each solid, set up the volume integral using the given base and cross-section shape, then evaluate.

1. The base of a solid is the region inside the circle $x^2 + y^2 = 4$. Cross sections perpendicular to the x-axis are squares. Find the volume.

$$V = \int_{-2}^2 (2\sqrt{4-x^2})^2 dx$$

Answer: _____

2. The base of a solid is the region inside the circle $x^2 + y^2 = 9$. Cross sections perpendicular to the x-axis are squares. Find the volume.

$$V = \int_{-3}^3 (2\sqrt{9-x^2})^2 dx$$

Answer: _____

3. The base of a solid is the region inside the circle $x^2 + y^2 = 1$. Cross sections perpendicular to the x-axis are squares. Find the volume.

$$V = \int_{-1}^1 (2\sqrt{1-x^2})^2 dx$$

Answer: _____

4. The base of a solid is the region inside the circle $x^2 + y^2 = 4$. Cross sections perpendicular to the x-axis are semicircles. Find the volume.

$$V = \int_{-2}^2 \frac{\pi}{2} (\sqrt{4-x^2})^2 dx$$

Answer: _____

5. The base of a solid is the region inside the circle $x^2 + y^2 = 4$. Cross sections perpendicular to the x-axis are equilateral triangles. Find the volume.

$$V = \int_{-2}^2 \frac{\sqrt{3}}{4} (2\sqrt{4-x^2})^2 dx$$

Answer: _____



6. The base of a solid is the region bounded by $y = x^2$ and $y = 4$. Cross sections perpendicular to the y -axis are squares. Find the volume.

$$V = \int_0^4 (2\sqrt{y})^2 dy$$

Answer: _____

7. The base of a solid is the region bounded by $y = x$ and $y = x^2$. Cross sections perpendicular to the x -axis are squares. Find the volume.

$$V = \int_0^1 (x - x^2)^2 dx$$

Answer: _____

8. The base of a solid is the region inside the ellipse $x^2/9 + y^2/4 = 1$. Cross sections perpendicular to the x -axis are squares. Find the volume.

$$V = \int_{-3}^3 \left(\frac{4}{3}\sqrt{9 - x^2}\right)^2 dx$$

Answer: _____

9. The base of a solid is the region inside the circle $x^2 + y^2 = 16$. Cross sections perpendicular to the x -axis are isosceles right triangles with the hypotenuse on the base. Find the volume.

$$V = \int_{-4}^4 \frac{1}{4} (2\sqrt{16 - x^2})^2 dx$$

Answer: _____

10. The base of a solid is the region bounded by $y = \sqrt{x}$, $y = 0$, and $x = 4$. Cross sections perpendicular to the x -axis are squares. Find the volume.

$$V = \int_0^4 (\sqrt{x})^2 dx$$

Answer: _____





Encourage students to sketch the base region and a representative cross section before writing the integral.

Solutions

1. The base of a solid is the region inside the circle $x^2 + y^2 = 4$. Cross sections perpendicular to the x-axis are squares. Find the volume.

$$V = \int_{-2}^2 (2\sqrt{4-x^2})^2 dx$$

- Solve the circle equation for y to get y equals plus or minus the square root of 4 minus x squared
- The side of the square equals the full vertical chord, which is 2 times the square root of 4 minus x squared
- Square the side to get the cross-sectional area, then integrate from negative 2 to 2
- Evaluate the integral to obtain 128 divided by 3

Answer: $\frac{128}{3}$

2. The base of a solid is the region inside the circle $x^2 + y^2 = 9$. Cross sections perpendicular to the x-axis are squares. Find the volume.

$$V = \int_{-3}^3 (2\sqrt{9-x^2})^2 dx$$

- Solve for y to get y equals plus or minus the square root of 9 minus x squared
- The square's side is 2 times the square root of 9 minus x squared
- Square the side and integrate from negative 3 to 3
- Evaluate to obtain 144

Answer: 144

3. The base of a solid is the region inside the circle $x^2 + y^2 = 1$. Cross sections perpendicular to the x-axis are squares. Find the volume.

$$V = \int_{-1}^1 (2\sqrt{1-x^2})^2 dx$$

- Solve for y to get y equals plus or minus the square root of 1 minus x squared
- The side of the square is 2 times the square root of 1 minus x squared
- Square the side and integrate from negative 1 to 1
- Evaluate to obtain 16 divided by 3

Answer: $\frac{16}{3}$

4. The base of a solid is the region inside the circle $x^2 + y^2 = 4$. Cross sections perpendicular to the x-axis are semicircles. Find the volume.

$$V = \int_{-2}^2 \frac{\pi}{2} (\sqrt{4-x^2})^2 dx$$

- The diameter of each semicircle equals the vertical chord, which is 2 times the square root of 4 minus x squared
- The radius is half of that, equal to the square root of 4 minus x squared
- The semicircle area is one half times pi times the radius squared
- Integrate from negative 2 to 2 to get 16 pi divided by 3

Answer: $\frac{16\pi}{3}$



5. The base of a solid is the region inside the circle $x^2 + y^2 = 4$. Cross sections perpendicular to the x-axis are equilateral triangles. Find the volume.

$$V = \int_{-2}^2 \frac{\sqrt{3}}{4} (2\sqrt{4-x^2})^2 dx$$

- The side of the equilateral triangle equals the full chord, which is 2 times the square root of 4 minus x squared
- The area of an equilateral triangle is the square root of 3 over 4 times the side squared
- Set up and integrate from negative 2 to 2
- Evaluate to obtain 32 times the square root of 3 divided by 3

Answer: $\frac{32\sqrt{3}}{3}$

6. The base of a solid is the region bounded by $y = x^2$ and $y = 4$. Cross sections perpendicular to the y-axis are squares. Find the volume.

$$V = \int_0^4 (2\sqrt{y})^2 dy$$

- Solve y equals x squared for x to get x equals plus or minus the square root of y
- The square has side length 2 times the square root of y
- Square the side and integrate from 0 to 4
- Evaluate to obtain 32

Answer: 32

7. The base of a solid is the region bounded by $y = x$ and $y = x^2$. Cross sections perpendicular to the x-axis are squares. Find the volume.

$$V = \int_0^1 (x - x^2)^2 dx$$

- Find the intersections of y equals x and y equals x squared, which are at x equals 0 and x equals 1
- The side of the square is the top function minus the bottom function, x minus x squared
- Square the side and integrate from 0 to 1
- Evaluate to obtain 1 divided by 30

Answer: $\frac{1}{30}$

8. The base of a solid is the region inside the ellipse $x^2/9 + y^2/4 = 1$. Cross sections perpendicular to the x-axis are squares. Find the volume.

$$V = \int_{-3}^3 \left(\frac{4}{3} \sqrt{9-x^2} \right)^2 dx$$

- Solve the ellipse equation for y to get y equals plus or minus two thirds times the square root of 9 minus x squared
- The side of the square is the full chord, four thirds times the square root of 9 minus x squared
- Square the side and integrate from negative 3 to 3
- Evaluate to obtain 64

Answer: 64



9. The base of a solid is the region inside the circle $x^2 + y^2 = 16$. Cross sections perpendicular to the x-axis are isosceles right triangles with the hypotenuse on the base. Find the volume.

$$V = \int_{-4}^4 \frac{1}{4} (2\sqrt{16 - x^2})^2 dx$$

→ The hypotenuse equals the full chord, 2 times the square root of 16 minus x squared

→ The area of an isosceles right triangle with hypotenuse h is h^2 divided by 4

→ Set up and integrate from negative 4 to 4

→ Evaluate to obtain 256 divided by 3

Answer: $\frac{256}{3}$

10. The base of a solid is the region bounded by $y = \sqrt{x}$, $y = 0$, and $x = 4$. Cross sections perpendicular to the x-axis are squares. Find the volume.

$$V = \int_0^4 (\sqrt{x})^2 dx$$

→ The side of each square equals the height of the region, which is the square root of x

→ Square the side to get x as the cross-sectional area

→ Integrate x from 0 to 4

→ Evaluate to obtain 8

Answer: 8

